CHAPTER THIRTEEN

EXERCISE 1

(a) (1) $S = \{1 ; 2 ; 3 ; 4 ; 5 ; 6\}$

(2) $A = \{1 ; 3 ; 5\}$ $B = \{1 ; 2 ; 3 ; 4\}$

(3) $n(A) = 3$

(4) $P(A) = \dfrac{n(A)}{n(S)} = \dfrac{3}{6} = \dfrac{1}{2}$

(5) $P(B) = \dfrac{n(B)}{n(S)} = \dfrac{4}{6} = \dfrac{2}{3}$

(b) (1) $S = \{A ; B ; C ; D ; E ; F ; G ; H\}$

(2) $n(X) = 5$

(3) $P(X) = \dfrac{n(X)}{n(S)} = \dfrac{5}{8}$

(4) $n(Y) = 2$

$\therefore P(Y) = \dfrac{n(Y)}{n(S)} = \dfrac{2}{8} = \dfrac{1}{4}$

(c) (1) $P(M) = \dfrac{n(M)}{n(S)}$ but $P(M) = \dfrac{1}{3}$ and $n(M) = 5$

$\therefore \dfrac{n(M)}{n(S)} = \dfrac{1}{3}$

$\therefore \dfrac{5}{n(S)} = \dfrac{1}{3}$

$\therefore n(S) = 15$

(2) $P(N) = \dfrac{n(N)}{n(S)} = \dfrac{3}{15} = \dfrac{1}{5}$

(3) $P(X) = \dfrac{n(X)}{n(S)}$ but $P(X) = 0,4 = \dfrac{2}{5}$

$\therefore \dfrac{n(X)}{n(S)} = \dfrac{2}{5}$

$\therefore \dfrac{n(X)}{15} = \dfrac{2}{5}$

$\therefore n(X) = 6$

(d) (1)(i) $n(S) = 50$

(ii) $n(E) = 10$

(2) $P(E) = \dfrac{n(E)}{n(S)} = \dfrac{10}{50} = \dfrac{1}{5}$

(3)(i) $P(\text{no one will win}) = \dfrac{20}{50} = \dfrac{2}{5}$

(ii) $\dfrac{10}{30} = \dfrac{1}{3}$

(e) (1)(i) $S = \{1 ; 2 ; 3 ; 4 ; 5 ; 6 ; 7 ; 8\}$

(ii) $P(\text{number less than 4}) = \dfrac{3}{8}$

(2)(i) $S_r = \{1 ; 3 ; 5 ; 7\}$

(ii) $P(\text{number less than 4}) = \dfrac{2}{4} = \dfrac{1}{2}$

(f)* (1) $S = \{HHH ; HHT ; HTH ; THH ; HTT ; THT ; TTH ; TTT\}$

(2)(i) $n(TTT) = 1$

$\therefore P(TTT) = \dfrac{n(TTT)}{n(S)} = \dfrac{1}{8}$

(ii) $n(\text{two heads}) = 3$

$\therefore P(\text{two heads}) = \dfrac{3}{8}$

(iii) $n(\text{at least two tails}) = 4$

$\therefore P(\text{at least two tails}) = \dfrac{4}{8} = \dfrac{1}{2}$

(3)(i) $S_r = \{THH ; THT ; TTH ; TTT\}$

(ii) $n(\text{at least two tails}) = 3$

$\therefore P(\text{at least two tails}) = \dfrac{3}{4}$

(g)* (1) $S = \{MM ; MF ; FM ; FF\}$

$\therefore P(FF) = \dfrac{n(FF)}{n(S)} = \dfrac{1}{4}$

(2) $S_r = \{FM ; FF\}$

$\therefore P(FF) = \dfrac{n(FF)}{n(S_r)} = \dfrac{1}{2}$

(3) $S_r = \{FM\ ;\ MF\ ;\ FF\}$

$\therefore P(FF) = \dfrac{n(FF)}{n(S_r)} = \dfrac{1}{3}$

(h) (1) $RF = \dfrac{f}{n} = \dfrac{5344}{24000} = 0{,}2227$

(2) Theoretical Probability: $P(Seven) = \dfrac{n(Seven)}{n(S)} = \dfrac{1}{12} = 0{,}0833$

$\therefore$ The relative frequency is not close to the theoretical probability. The die is not fair.

(3) $E = \{1\ ;\ 2\ ;\ 3\ ;\ 4\ ;\ 5\}$

$S = \{1\ ;\ 2\ ;\ 3\ ;\ 4\ ;\ 5\ ;\ 6\ ;\ 7\ ;\ 8\ ;\ 9\ ;\ 10\ ;\ 11\ ;\ 12\}$

Theoretical Probability: $P(E) = \dfrac{n(E)}{n(S)} = \dfrac{5}{12}$

We expect the relative frequency to be close to the theoretical probability:

$RF = \dfrac{f}{n} = \dfrac{f}{24000} \approx \dfrac{5}{12}$

$\therefore f \approx \dfrac{5}{12} \times 24\,000 = 10\,000$

We expect E to take place approximately 10 000 times.

(i) (1) $n(S) = 52$

$n(Hearts) = 13$

Theoretical Probability: $P(Hearts) = \dfrac{n(Hearts)}{n(S)} = \dfrac{13}{52} = \dfrac{1}{4}$

We expect the relative frequency to be close to the theoretical probability:

$RF = \dfrac{f}{n} = \dfrac{f}{50\,000} \approx \dfrac{1}{4}$

$\therefore f \approx \dfrac{1}{4} \times 50\,000 = 12\,500$

Most reasonable number of times is C (12 685 times).

(2) $RF = \dfrac{f}{n} = \dfrac{f}{72\,000} = \dfrac{1}{4}$

$\therefore f \approx \dfrac{1}{4} \times 72\,000 = 18\,000$

We expect that a card of hearts will be selected approximately 18 000 times.

(j)* (1) $RF = \dfrac{f}{n} = \dfrac{80\,000}{120\,000} = \dfrac{2}{3}$

(2) $n(S) = 800$

$P(X) = \dfrac{n(X)}{n(S)}$ and $RF = \dfrac{2}{3}$

$\therefore \dfrac{n(X)}{800} \approx \dfrac{2}{3}$

$\therefore n(X) \approx \dfrac{2}{3} \times 800 \approx 533$ times

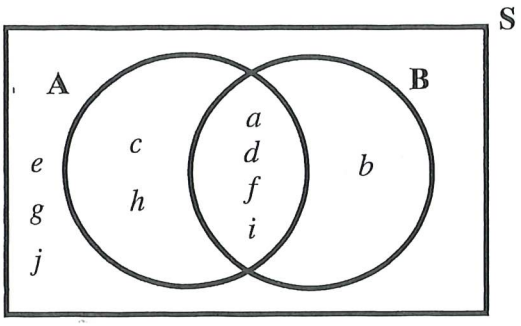
(3) $\dfrac{f}{33000} \approx \dfrac{2}{3}$

$\therefore f \approx \dfrac{2}{3} \times 33\,000 \approx 22\,000$ times

EXERCISE 2

(a) $S = \{a\ ;\ b\ ;\ c\ ;\ d\ ;\ e\ ;\ f\ ;\ g\ ;\ h\ ;\ i\ ;\ j\}$
$A = \{a\ ;\ c\ ;\ d\ ;\ f\ ;\ h\ ;\ i\}$
$B = \{a\ ;\ b\ ;\ d\ ;\ f\ ;\ i\}$

(1)



(2)(i) not A $= \{b\ ;\ e\ ;\ g\ ;\ j\}$

(ii) not B $= \{c\ ;\ e\ ;\ g\ ;\ h\ ;\ j\}$

(iii) A and B $= \{a\ ;\ d\ ;\ f\ ;\ i\}$

(iv) A or B $= \{a\ ;\ b\ ;\ c\ ;\ d\ ;\ f\ ;\ h\ ;\ i\}$

(v) not(A and B) $= \{b\ ;\ c\ ;\ e\ ;\ g\ ;\ h\ ;\ j\}$

(vi) not(A or B) $= \{e\ ;\ g\ ;\ j\}$

(3)(i) $n(\text{A}) = 6$

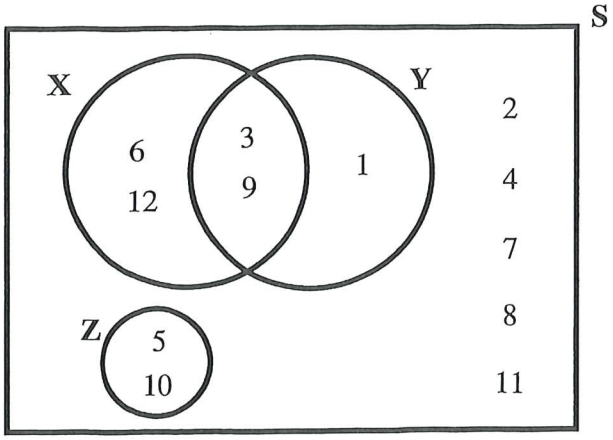
(ii) $\text{P(not B)} = \dfrac{n(\text{not B})}{n(\text{S})} = \dfrac{5}{10} = \dfrac{1}{2}$

(iii) $\text{P(A and B)} = \dfrac{n(\text{A and B})}{n(\text{S})} = \dfrac{4}{10} = \dfrac{2}{5}$

(iv) $n\left[\text{not(A or B)}\right] = 3$

(b) $X = \{3\ ;6\ ;9\ ;12\}$
$Y = \{1\ ;3\ ;9\}$
$Z = \{5\ ;10\}$

(1)



(2)(i) $n(\text{X}) = 4$

(ii) $\text{P(Y)} = \dfrac{n(\text{Y})}{n(\text{S})} = \dfrac{3}{12} = \dfrac{1}{4}$

(iii) $\text{P(Z)} = \dfrac{n(\text{Z})}{n(\text{S})} = \dfrac{2}{12} = \dfrac{1}{6}$

(iv) $\text{P(not X)} = \dfrac{n(\text{not X})}{n(\text{S})} = \dfrac{8}{12} = \dfrac{2}{3}$

(v) $\text{P(X and Y)} = \dfrac{n(\text{X and Y})}{n(\text{S})} = \dfrac{2}{12} = \dfrac{1}{6}$

(vi) $\text{P(Y and Z)} = \dfrac{n(\text{Y and Z})}{n(\text{S})} = \dfrac{0}{12} = 0$

(vii) $\text{P(X or Y)} = \dfrac{n(\text{X or Y})}{n(\text{S})} = \dfrac{5}{12}$

(viii) $n(\text{X or Z}) = 6$

(ix) $\text{P(not(Y or Z))} = \dfrac{n(\text{not(Y or Z))}}{n(\text{S})} = \dfrac{7}{12}$

(x) $P(\text{not(X and Z)}) = \dfrac{n(\text{not(X and Z))}}{n(\text{S})} = \dfrac{12}{12} = 1$

(c) (1) $10 + 30 + 20 + 90 = 150$ Grade 10 learners

(2)(i) $\text{P(A)} = \dfrac{n(\text{A})}{n(\text{S})} = \dfrac{50}{150} = \dfrac{1}{3}$

(ii) $n(\text{not G}) = 110$

(iii) $\text{P(G and A)} = \dfrac{n(\text{G and A})}{n(\text{S})} = \dfrac{30}{150} = \dfrac{1}{5}$

(iv) $n(\text{G or A}) = 60$

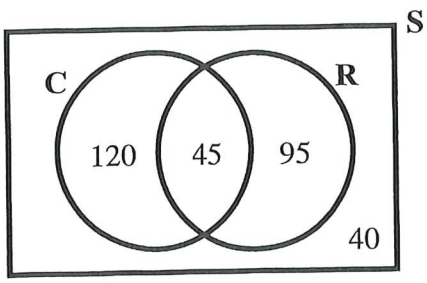
(v) $\text{P(not (G and A))} = \dfrac{n(\text{not (G and A))}}{n(\text{S})} = \dfrac{120}{150} = \dfrac{4}{5}$

(3) 90

(4) $\text{P(at least one of the subjects)} = \dfrac{10 + 20 + 30}{150} = \dfrac{60}{150} = \dfrac{2}{5}$

(d) $n(\text{S}) = 300$
$n(\text{C}) = 165$
$n(\text{C and R}) = 45$

(1)



(2)(i) $n(\text{R}) = 140$

(ii) $\text{P(not R)} = = \dfrac{n(\text{not R})}{n(\text{S})} = \dfrac{160}{300} = \dfrac{8}{15}$

(iii) $\text{P(C and R)} = \dfrac{n(\text{C and R})}{n(\text{S})} = \dfrac{45}{300} = \dfrac{3}{20}$

(iv) $\text{P(C or R)} = \dfrac{n(\text{C or R})}{n(\text{S})} = \dfrac{260}{300} = \dfrac{13}{15}$

(v) $n(\text{not (C and R)}) = 120 + 95 + 40 = 255$

(vi) $\text{P(not (C or R))} = \dfrac{n(\text{not (C or R)})}{n(\text{S})} = \dfrac{40}{300} = \dfrac{2}{15}$

(e) (1) $0{,}3 + 0{,}1 + 0{,}4 + q = 1$

$\therefore q = 1 - 0{,}8$

$\therefore q = 0{,}2$

(2)(i) $\text{P(A)} = 0{,}3 + 0{,}1 = 0{,}4$

(ii) $\text{P(B)} = 0{,}1 + 0{,}4 = 0{,}5$

(iii) $\text{P(not A)} = 0{,}4 + 0{,}2 = 0{,}6$

(iv) $\text{P(not B)} = 0{,}3 + 0{,}2 = 0{,}5$

(v) $\text{P(A and B)} = 0{,}1$

(vi) $\text{P(A or B)} = 0{,}3 + 0{,}1 + 0{,}4 = 0{,}8$

(3) $n(\text{A}) = 12$

(i) $\text{P(A or B)} = 0{,}8$ and $\text{P(A)} = 0{,}4$

but $\text{P(A or B)} = \dfrac{n(\text{A or B})}{n(\text{S})}$ and $\text{P(A)} = \dfrac{n(\text{A})}{n(\text{S})}$

$\therefore 0{,}8 = \dfrac{n(\text{A or B})}{n(\text{S})}$ $\therefore 0{,}4 = \dfrac{12}{n(\text{S})}$

$\therefore 0{,}8 = \dfrac{n(\text{A or B})}{n(30)}$ $\therefore n(\text{S}) = \dfrac{12}{0{,}4}$

$\therefore n(\text{A or B}) = 0{,}8 \times 30$ $\therefore n(\text{S}) = 30$

$\therefore n(\text{A or B}) = 24$

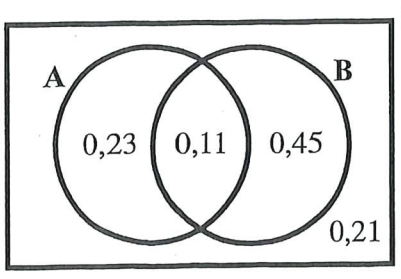
(ii) $\text{P(A and B)} = \dfrac{n(\text{A and B})}{n(\text{S})}$

$\therefore 0{,}1 = \dfrac{n(\text{A and B})}{30}$

$\therefore n(\text{A and B}) = 30 \times 0{,}1 = 3$

$\therefore n(\text{not(A and B)}) = 30 - 3 = 27$

(f) $\text{P(B)} = 0{,}56$; $\text{P(A and B)} = 0{,}11$; $\text{P(A or B)} = 0{,}79$



(1) $P(A) = 0,23 + 0,11 = 0,34$

(2) $P(\text{not(A and B)}) = 0,23 + 0,45 + 0,21 = 0,89$

(3) $n(\text{not(A or B)})$ and $n(S) = 400$

$$P(\text{A or B}) = \frac{n(\text{A or B})}{n(S)} \quad \text{but} \quad P(\text{A or B}) = 0,79$$

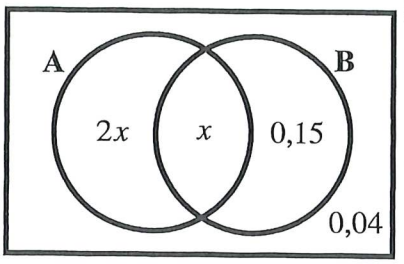
$$\therefore \frac{n(\text{A or B})}{400} = 0,79$$

$\therefore n(\text{A or B}) = 0,79 \times 400 = 316$

$\therefore n(\text{not(A or B)}) = 400 - 316$

$\therefore n(\text{not(A or B)}) = 84$

(g) (1)



$P(\text{A and B}) = x$
and $P(\text{A}) = 3x$

(2) $2x + x + 0,15 + 0,04 = 1$

$\therefore 3x = 1 - 0,15 - 0,04$

$\therefore 3x = 0,81$

$\therefore x = 0,27$

(3) (i) $P(\text{A}) = 3x$ and $x = 0,27$

$\therefore P(\text{A}) = 3(0,27)$

$\therefore P(\text{A}) = 0,81$

(ii) $P(\text{B}) = x + 0,15$

$\therefore P(\text{B}) = 0,27 + 0,15$

$\therefore P(\text{B}) = 0,42$

(iii) $P(\text{not(A and B)}) = 2x + 0,15 + 0,04$

$\therefore P(\text{not(A and B)}) = 2(0,27) + 0,15 + 0,04$

$\therefore P(\text{not(A and B)}) = 0,73$

(h)* $P(\text{Y}) = \frac{1}{2}$ and $P(\text{Shaded area}) = \frac{2}{3}$

(1) $P(\text{not(X or Y)}) = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$

$\therefore P(\text{X or Y}) = 1 - \frac{1}{6} = \frac{5}{6}$

(2) $n(X) = 21$ and $n(Y) = 14$

$$P(Y) = \frac{n(Y)}{n(S)}$$

$$\therefore \frac{1}{2} = \frac{14}{n(S)}$$

$$\therefore n(S) = 28$$

$$P(X) = \frac{n(X)}{n(S)} = \frac{21}{28} = \frac{3}{4} \quad \text{and} \quad P(\text{unshaded area}) = 1 - \frac{2}{3} = \frac{1}{3}$$

$$\therefore P(X \text{ and } Y) = P(X) - P(\text{unshaded area})$$

$$= \frac{3}{4} - \frac{1}{3}$$

$$= \frac{9-4}{12}$$

$$= \frac{5}{12}$$

EXERCISE 3

(a) $P(A) = 0,6$; $P(B) = 0,3$; $P(A \text{ and } B) = 0,2$

(1) $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$\therefore P(A \text{ or } B) = 0,6 + 0,3 - 0,2$

$\therefore P(A \text{ or } B) = 0,7$

(2) $P(\text{not } A) = 1 - P(A)$

$\therefore P(\text{not } A) = 1 - 0,6$

$\therefore P(\text{not } A) = 0,4$

(3) $P(\text{not } B) = 1 - P(B)$

$\therefore P(\text{not } B) = 1 - 0,3$

$\therefore P(\text{not } B) = 0,7$

(4) $P(\text{not}(A \text{ or } B)) = 1 - P(A \text{ or } B)$

$\therefore P(\text{not}(A \text{ or } B)) = 1 - 0,7$

$\therefore P(\text{not}(A \text{ or } B)) = 0,3$

(b) $P(X) = 0,5$; $P(Y) = 0,6$; $P(X \text{ or } Y) = 0,9$

(1)(i) $P(\text{not } Y) = 1 - P(Y)$

$\therefore P(\text{not } Y) = 1 - 0,6 = 0,4$

(ii) $P(\text{not}(X \text{ or } Y)) = 1 - P(X \text{ or } Y)$

$\therefore P(\text{not}(X \text{ or } Y)) = 1 - 0,9 = 0,1$

(iii) $P(X \text{ or } Y) = P(X) + P(Y) - P(X \text{ and } Y)$

$\therefore 0,9 = 0,5 + 0,6 - P(X \text{ and } Y)$

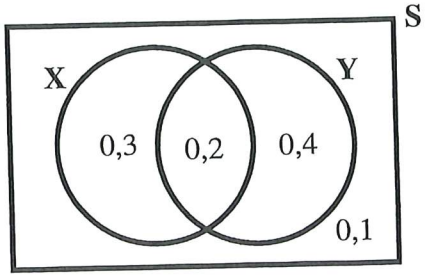
$\therefore P(X \text{ and } Y) = 0,5 + 0,6 - 0,9$

$\therefore P(X \text{ and } Y) = 0,2$

(iv) $P(\text{not}(X \text{ and } Y)) = 1 - P(X \text{ and } Y)$

$\therefore P(\text{not}(X \text{ and } Y)) = 1 - 0,2 = 0,8$

(2)



(c) $P(M) = 0,35$; $P(M \text{ and } N) = 0,2$; $P(M \text{ or } N) = 0,85$

(1) $P(\text{not } M) = 1 - P(M)$

$\therefore P(\text{not } M) = 1 - 0,35 = 0,65$

(2) $P(M \text{ or } N) = P(M) + P(N) - P(M \text{ and } N)$

$\therefore 0,85 = 0,35 + P(N) - 0,2$

$\therefore P(N) = 0,85 - 0,35 + 0,2$

$\therefore P(N) = 0,7$

$P(\text{not } N) = 1 - P(N)$

$\therefore P(\text{not } N) = 1 - 0,7 = 0,3$

(d) $P(A) = \frac{1}{3}$; $P(\text{not } B) = \frac{3}{4}$; $P(A \text{ and } B) = \frac{1}{6}$

(1)(i) $P(B) = 1 - P(\text{not } B)$

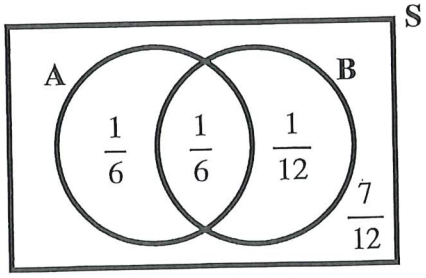
$\therefore P(B) = 1 - \frac{3}{4} = \frac{1}{4}$

(ii) $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

$\therefore P(A \text{ or } B) = \frac{1}{3} + \frac{1}{4} - \frac{1}{6}$

$\therefore P(A \text{ or } B) = \frac{4+3-2}{12} = \frac{5}{12}$

(2)



(e) $P(Q) = 0,5$; $P(R) = 0,3$; $P(\text{not}(Q \text{ or } R)) = 0,4$

(1) $P(\text{not}(Q \text{ or } R)) = 1 - P(Q \text{ or } R)$

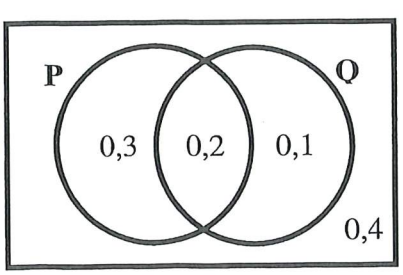
$\therefore P(Q \text{ or } R) = 1 - 0,4 = 0,6$

$P(Q \text{ or } R) = P(Q) + P(R) - P(Q \text{ and } R)$

$\therefore 0,6 = 0,5 + 0,3 - P(Q \text{ and } R)$

$\therefore P(Q \text{ and } R) = 0,5 + 0,3 - 0,6 = 0,2$

(2)



(f) $P(\text{not } X) = \frac{5}{8}$; $P(X \text{ or } Y) = \frac{7}{8}$; $P(\text{not}(X \text{ and } Y)) = \frac{3}{4}$

(1) $P(X) = 1 - P(\text{not } X)$

$\therefore P(X) = 1 - \frac{5}{8} = \frac{3}{8}$

$P(X \text{ and } Y) = 1 - P(\text{not}(X \text{ and } Y))$

$\therefore P(X \text{ and } Y) = 1 - \frac{3}{4}$

$\therefore P(X \text{ and } Y) = \frac{1}{4}$

$P(X \text{ or } Y) = P(X) + P(Y) - P(X \text{ and } Y)$

$\therefore \frac{7}{8} = \frac{3}{8} + P(Y) - \frac{1}{4}$

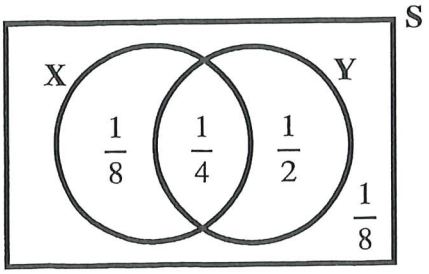
$\therefore P(Y) = \frac{7}{8} - \frac{3}{8} + \frac{1}{4}$

$\therefore P(Y) = \frac{6}{8} = \frac{3}{4}$

$P(\text{not } Y) = 1 - P(Y)$

$\therefore P(\text{not } Y) = 1 - \frac{3}{4} = \frac{1}{4}$

(2)



(g) $P(A) = 5x$; $P(B) = 4x$; $P(\text{not}(A \text{ and } B)) = 0,9$ and $P(A \text{ or } B) = 0,8$

(1) $P(A \text{ and } B) = 1 - P(\text{not}(A \text{ and } B))$
$\therefore P(A \text{ and } B) = 1 - 0,9$
$\therefore P(A \text{ and } B) = 0,1$

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
$\therefore 0,8 = 5x + 4x - 0,1$
$\therefore 0,9 = 9x$
$\therefore x = 0,1$

(2) $P(\text{not } B) = 1 - P(B)$
$\therefore P(\text{not } B) = 1 - (4 \times (0,1))$
$\therefore P(\text{not } B) = 0,6$

(h) $P(X \text{ and } Y) = 0,3$; $P(\text{not}(X \text{ or } Y)) = 0,1$ and $P(Y) = 2 \cdot P(X)$

$P(X \text{ or } Y) = 1 - P(\text{not}(X \text{ or } Y))$
$\therefore P(X \text{ or } Y) = 1 - 0,1$
$\therefore P(X \text{ or } Y) = 0,9$

$P(X \text{ or } Y) = P(X) + P(Y) - P(X \text{ and } Y)$
$\therefore 0,9 = P(X) + 2\,P(X) - 0,3$
$\therefore 1,2 = 3\,P(X)$
$\therefore P(X) = 0,4$

(i) $P(A) = 0,92$; $P(B) = 0,98$; $P(A \text{ and } B) = 0,94$

$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
$\therefore P(A \text{ or } B) = 0,92 + 0,98 - 0,94$
$\therefore P(A \text{ or } B) = 0,96$

$P(\text{not}(A \text{ or } B)) = 1 - P(A \text{ or } B)$
$\therefore P(\text{not}(A \text{ or } B)) = 1 - 0,96$
$\therefore P(\text{not}(A \text{ or } B)) = 0,04$

$\therefore$ The probability that smoke will not be detected is 0,04.

(j) (1) $n(\text{Laptop or Tablet}) = 50 - 17 = 33$

(2) $n(\text{L or T}) = n(\text{L}) + n(\text{T}) - n(\text{L and T})$
$\therefore 33 = 21 + 25 - n(\text{L and T})$
$\therefore n(\text{L and T}) = 46 - 33 = 13$
$\therefore$ 13 people own both devices.

EXERCISE 4

(a) $P(A) = 0,2$ and $P(B) = 0,4$

(1) $P(A \text{ and } B) = 0$

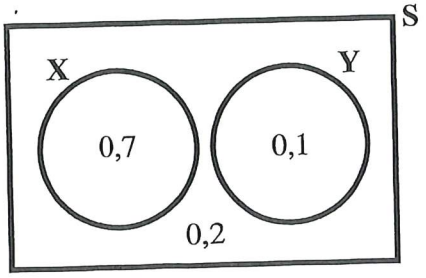
(2) $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
$\therefore P(A \text{ or } B) = 0,2 + 0,4 - 0$
$\therefore P(A \text{ or } B) = 0,6$

(b) $P(\text{not } X) = 0,3$ and $P(X \text{ or } Y) = 0,8$

(1) $P(X) = 1 - P(\text{not } X)$
$\therefore P(X) = 1 - 0,3$
$\therefore P(X) = 0,7$

$P(X \text{ or } Y) = P(X) + P(Y) - P(X \text{ and } Y)$
$\therefore 0,8 = 0,7 + P(Y) - 0$
$\therefore P(Y) = 0,1$

(2)



(c) $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{3}$

(1) $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
$\therefore P(A \text{ or } B) = \frac{1}{2} + \frac{1}{3} - \frac{1}{6}$
$\therefore P(A \text{ or } B) = \frac{3 + 2 - 1}{6}$
$\therefore P(A \text{ or } B) = \frac{4}{6} = \frac{2}{3}$

(2) $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
$\therefore P(A \text{ or } B) = \frac{1}{2} + \frac{1}{3} - 0$
$\therefore P(A \text{ or } B) = \frac{5}{6}$

(d) $P(M \text{ or } N) = 0,6$ and $P(\text{not } N) = 0,65$

(1) $P(N) = 1 - P(\text{not } N)$
$\therefore P(N) = 1 - 0,65$
$\therefore P(N) = 0,35$

$P(M \text{ or } N) = P(M) + P(N) - P(M \text{ and } N)$
$\therefore 0,6 = P(M) + 0,35 - 0$
$\therefore P(M) = 0,25$

(2) $P(M \text{ and } N) = 1 - P(\text{not } (M \text{ and } N))$
$\therefore P(M \text{ and } N) = 1 - \frac{4}{5}$
$\therefore P(M \text{ and } N) = \frac{1}{5} = 0,2$

$P(M \text{ or } N) = P(M) + P(N) - P(M \text{ and } N)$
$\therefore 0,6 = P(M) + 0,35 - 0,2$
$\therefore P(M) = 0,6 - 0,35 + 0,2$
$P(M) = 0,45$

(e) $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
$\therefore 1 = 0,32 + P(B) - 0$
$\therefore P(B) = 1 - 0,32$
$\therefore P(B) = 0,68$

$P(\text{not } B) = 1 - P(B)$
$\therefore P(\text{not } B) = 1 - 0,68 = 0,32$

(f) $P(X) = 0,3$ and $P(Y) = 0,7$

(1) $P(X \text{ and } Y) = 0$

(2) $P(X \text{ or } Y) = 1 - P(\text{not}(X \text{ or } Y))$
$\therefore P(X \text{ or } Y) = 1 - 0,2$
$\therefore P(X \text{ or } Y) = 0,8$

$P(X \text{ or } Y) = P(X) + P(Y) - P(X \text{ and } Y)$
$\therefore 0,8 = 0,3 + 0,7 - P(X \text{ and } Y)$
$\therefore P(X \text{ and } Y) = 0,3 + 0,7 - 0,8$
$\therefore P(X \text{ and } Y) = 0,2$

(g) $P(A) = 0,7$ and $P(B) = 0,6$
$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
$\therefore 1 = 0,7 + 0,6 - P(A \text{ and } B)$
$\therefore P(A \text{ and } B) = 0,7 + 0,6 - 1$
$\therefore P(A \text{ and } B) = 0,3$

(h) (1)(i) N and T (ii) M and N

(2)(i) $P(M \text{ and } N) = \frac{2}{8} = \frac{1}{4}$

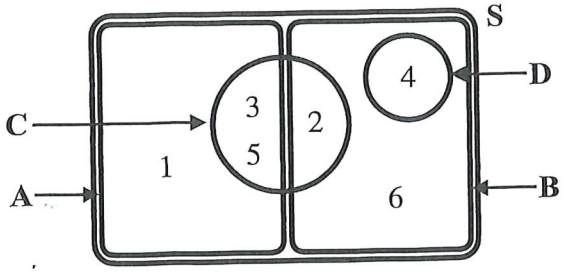
(ii) $P(N \text{ and } T) = 0$

(iii) $P(N \text{ or } T) = \frac{7}{8}$

(iv) $P(M \text{ or } N) = 1$

(i) $A = \{1;\ 3;\ 5\};\ B = \{2;\ 4;\ 6\};\ C = \{2;\ 3;\ 5\};\ D = \{4\}$

(1)



(2)(i) A and B
A and D
C and D

(ii) A and B

(iii) A and B

(3)(i) $P(A \text{ and } C) = \frac{2}{6} = \frac{1}{3}$

(ii) $P(A \text{ and } B) = 0$

(iii) $P(A \text{ or } B) = 1$

(iv) $P(C \text{ or } D) = \frac{4}{6} = \frac{2}{3}$

(j) (1)(i) $P(R) = \frac{100}{240} = \frac{5}{12}$

(ii) $P(G) = \frac{120}{240} = \frac{1}{2}$

(iii) $P(L) = \frac{140}{240} = \frac{7}{12}$

(iv) $P(U) = \frac{120}{240} = \frac{1}{2}$